NOT MEASUREMENT SENSITIVE

MIL-S-83825
22 FEBRUARY 1992

MILITARY SPECIFICATION

SWEEPER, STREET AND AIRFIELD, VACUUM OR REGENERATIVE TYPE, DIESEL ENGINE DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This purchase description covers a self propelled, vacuum or recirculating (regenerative) air type street and airfield sweepers.

1.2 Classification. The vehicle shall be one of the sizes shown in table I, as specified (see 6.2).

TABLE I. Sweeper Classification.

Size	Minimum Useable Hopper Capacity (Cubic Yards)	Minimum Gross Vehicle Weight Rating (Pounds)
I	3.5	19,000
II	6.0	24,000

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Warner Robins Air Logistics Center, ATTN: WR-ALC/LVRG, Robins AFB GA 31098-5345, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-3825

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

KKK-T-2107	- Trucks and Truck Tractors: Commercial, Diesel or Gasoline Engine Driven, 14,000 to 21,000 Pounds GVW, 4x2.
KKK-T-2108	- Trucks and Truck Tractors: Commercial, Diesel or Gasoline Engine Driven, 24,000 to 32,000 Pounds GVW, 4x2.

MILITARY

MIL-T-12459	- Tire, Pneumatic: for Military Ground Vehicles.
MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-831	- Test Reports, Preparation of.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-965	- Parts Control Program.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.
MIL-STD-1556	- Government/Industry Data Exchange Program Contractor Participation Requirements.
MIL-STD-1791	- Designing for Internal Aerial Delivery in Fixed Wing Aircraft.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of invitation for bid or request for proposal.

US AIR FORCE (USAF) DRAWINGS

8838302	- Fastener Quality Assurance Requirements for WR-ALC/MMV Vehicle and Support Equipment.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of documents shall be those in effect on the date of invitation for bid or request for proposal.

AMERICAN WELDING SOCIETY (AWS)

B2.1-84 - Welding Procedure and Performance Qualification.

(Application for copies should be addressed to American Welding Society, 2501 NW Seventh Street, Miami FL 33125.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J366 - Exterior Sound Level for Heavy Trucks and Buses.
J534 - Lubrication Fittings.

(Application for copies should be addressed to Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale PA 15096.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Design.

3.1.1 Materials. Materials shall be as specified herein. Materials not specifically covered by this specification or referenced document shall be suitable in every respect for this type equipment. Except as otherwise specified, wood shall not be used.

3.1.1.1 Protective treatment. Materials that are subject to deterioration shall be protected in such a manner that shall in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip, crack, or scale with age or extremes of climatic conditions shall not be used.

3.1.1.2 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.1.2 Construction. The sweeper shall be a vacuum or regenerative air type capable of sweeping streets, curbs and gutters, runways, taxiways, ramps and parking areas. The Size I sweeping unit shall be mounted on a Type I, Class C diesel powered truck chassis as specified in KKK-T-2107. The Size II sweeping unit shall be mounted on a Type I, Class A diesel powered truck chassis a specified in KKK-T-2108. The unit shall, as a minimum, include a diesel engine, power train, fan, pickup head, material separator, dust control system, and a hydraulically operated dumping hopper. The unit shall operate in dry or wet weather without harming the filtration system. The sweeper shall be constructed so that parts do not work loose in service and shall withstand the loads,

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shocks, vibrations, and other conditions incident to operation, shipping, and storage with minimum loss of time for maintenance, repair, and servicing. Its design shall ensure safe, efficient, and economical operation. All assemblies, controls, and installed equipment shall be located so that there is no adverse interference with each other, or with the operation, and shall be readily accessible for maintenance, operation, and replacement.

3.1.2.1 Commercial product. The sweeper and attachments must be the manufacturer's standard commerical product (see 6.6), except as modified to meet the requirements of this specification. Upon request of the contracting officer, offerer must submit evidence to establish commerciality of the sweeper and furnish names and addresses of non-Government sources that have acquired the sweeper at least a year prior to the solicitation opening date.

3.1.2.2 Maintainability. The vehicle shall be designed and constructed to provide the following:

a. A minimum number of parts consistent with the performance required herein.

b. A minimum amount of time for assembling, disassembling, locating trouble sources, and maintaining the vehicle. Where practical, components shall be located for rapid inspection and recognition of potential failure.

c. Ease of adjusting, servicing, and replacing parts and components.

d. Use of readily available standard tools and equipment for maintenance.

e. Maintenance with a minimum number of tools.

3.1.2.3 Foolproofness. Where improper installation of an item could cause a malfunction, an asymmetric mounting system shall be provided, where practical, to ensure proper mounting of the item.

3.1.2.4 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers, or nylon patches shall not be used for this purpose, except for attachment of trim items or as provided in commercial (see 6.6) components. Tapped aluminum threads shall have a minimum thread engagement of at least two times the nominal fastener diameter. All fasteners of Grade 5 or higher must meet the quality assurance requirements of US Air Force Drawing 8838302.

3.1.2.5 Servicing provisions. Drains, grease fittings, lines, and check points for all components shall be so located that they are readily accessible and do not require special tools for proper servicing.

3.1.2.6 Lubrication. All parts normally requiring grease for lubrication shall be provided with lubrication fittings in accordance with SAE J534. Where the use of high lubrication pressures may damage grease seals, a pressure relief device shall be provided. Extended fittings shall be provided to lubricate parts or assemblies which are not readily accessible for direct lubrication or which are likely to be overlooked because of inaccessibility.

3.1.2.7 Component protection. All space in which work is performed during operation, service, and maintenance of the vehicle shall be free of hazardous protrusions, sharp edges, or other features which may cause injury to personnel.

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All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard.

3.1.2.8 Foreign object damage. All loose metal parts, such as pins and valve caps, shall be securely attached so they cannot become separated from the vehicle.

3.1.2.9 Parts control. The contractor shall participate in the Government Industry Data Exchange Program (GIDEP) in accordance with MIL-STD-1556. In the design of the vehicle, the contractor shall ensure the components selected have not been identified as a problem or unresolved deficiency by GIDEP. The contractor shall submit an Alert/Safe-Alert on a DD Form 1939 whenever an actual or potential problem is evident with respect to a part, component, material, process, safety condition or to test equipment. The contractor shall certify maximum use of off-the-shelf items in accordance with MIL-STD-965.

3.1.3 Temperature conditions. The sweeper shall be capable of continual performance at any ambient temperature from -20°F to +125°F when serviced in accordance with the supplier's recommendations and with all water from dust suppression system drained when operated in temperatures below freezing (+32°F).

3.1.4 Weight distribution. The weight distribution of the vehicle in any loading condition shall not be unbalanced or improperly distributed. The difference in load between the left and right wheels of any axle shall not exceed 2.5 percent of the total load on the axle.

3.1.5 Electromagnetic interference. The requirements of KKK-T-2107 and KKK-T-2108 for electromagnetic radiation suppression shall apply not only to the truck chassis but also to the entire vehicle (i.e. they shall apply after the sweeping unit and auxiliary engine have been installed on the truck chassis).

3.1.6 Sound levels.

3.1.6.1 Cab interior sound level. The cab interior sound level shall not exceed 84 dB(A) when the vehicle is tested in accordance with 4.6.5.4.1.

3.1.6.2 Exterior sound level. The exterior sound level shall not exceed 88 dB(A) when measured in accordance with 4.6.5.4.2. If the exterior sound level exceeds 84 dB(A), a noise hazard caution sign(s) shall be posted in accordance with paragraph 4.3 of MIL-STD-1474.

3.1.7 Air transportability. The vehicle shall be air transportable without shoring in accordance with MIL-STD-1791 in Types C-130, C-141, and C-5 Aircraft and shall be air transportable in a Type C-17 Aircraft. Preliminary design guidelines for the Type C-17 are 142 inches maximum height, 204 inches maximum width, 812 inches maximum length, 90 ramp angle, and floor load capacity equal to that allowed for the Type C-5 Aircraft.

3.1.7.1 Equipment removal. Removal of equipment shall be permitted, if necessary, to provide the required height and width for air transport. Removal and reinstallation of equipment shall be limited to removal and attachment of standard fasteners and fittings with common hand tools. The weight of individual parts or assemblies which must be handled or positioned shall not exceed the limits specified in MIL-STD-1472 for a one person lift. All equipment removed shall be stored on the vehicle; all necessary caps and plugs

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shall be provided to permit driving and storage of the vehicle while in the air transport configuration. Removal or reinstallation of the equipment shall not take longer than 15 minutes for two mechanics using common hand tools.

3.1.7.2 Tiedowns. The vehicle shall be equipped with sufficient tiedown points so located that it can be restrained in all directions during air transport. The attachment points shall be designed for use with attachment in the aircraft as described in MIL-STD-1791. Sufficient clearance shall be maintained around all attachment points to allow for restraining chains without interference from any other structure. The tiedown points shall have a minimum clear opening of three inches to allow attachment of a chain with a grabhook.

3.2 Performance.

3.2.1 Mobility. The fully loaded and equipped sweeper shall attain the following performance on dry, paved surfaces:

a. Highway travel at a speed not less than 50 miles per hour (mph).

b. An outside vehicle turning radius not greater than 25 feet to the right and to the left.

c. A cornering radius not greater than 12 feet.

d. An overall ground clearance, when all the accessories and attachments are in the travel position, of not less than 5 inches.

3.2.2 Cleaning ability.

3.2.2.1 Sand pickup. The sweeper shall pick up commercially procured clean dry sand from a paved surface covered with a density spread of 0.25 pounds per square foot. Pick up requirements shall be not less than those specified in Table II below:

TABLE II - SAND PICKUP

Speed (mph)	Density of Spread (Pounds per Sq Ft)	Spread Area (Sq Ft)	Pickup (Percent)
3 minimum	0.25	300	95
6 minimum	0.25	300	85

3.2.2.2 Pea gravel. The sweeper shall pick up pea gravel at a rate not less than that specified in Table III below:

TABLE III - PEA GRAVEL PICKUP

Speed (mph)	Density of Spread (Pounds per Sq Ft)	Spread Area (Sq Ft)	Pickup (Percent)
3 minimum	0.25	300	95
6 minimum	0.25	300	95

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Gradation of the pea gravel shall be such that 100 percent passes a 1/2 inch screen and not more than 2 percent passes a standard US No. 8 screen.

3.2.2.3 Stone pickup. While traveling at a speed of 3 mph minimum, the sweeper shall pick up and retain not less than 18 out of 20 stones measuring at least 2 inches in diameter and passing through a 3 inch sieve. Stones shall have been placed on a paved surface approximately 36 inches apart.

3.2.2.4 Street pickup. While traveling at a speed of at least 3 mph, the sweeper shall pick up and retain 100 percent of general trash such as paper, plastic cups, rolled newspapers, soft drink cans, 1 liter soft brink bottles, and tree limbs at least 13 inches long and 1/2 inch diameter which has been scattered in the gutterline of the street for a distance of 300 ft. The street to be negotiated shall be curbed and shall include curves having radii of not less than 15 feet.

3.2.2.5 Hand suction hose. The hand suction hose and nozzle shall have an 8 foot minimum reach on either the right or left side of the sweeper and shall have a suction sufficient to pick up and retain items in, 3.2.2.4 and particles ranging in size from fine dust (10 microns) to 2 1/2 inch diameter stones.

3.2.2.6 Dust emission. During normal wind conditions and normal cleaning operations, dust created by the action of the gutter brooms shall be reduced by a dust suppression system. For sweeping units designed to exhaust to the surrounding air, the capacity of the emissions shall be such that there is no dust visible to a person 50 feet from the sweeper during the performance of cleaning requirements 3.2.2.1 thru 3.2.2.5.

3.3 Chassis.

3.3.1 Truck chassis. The size I sweeping unit shall be mounted on a truck chassis in accordance with KKK-T-2107, Type I, Class C. The size II sweeping unit shall be mounted on a truck chassis in accordance with KKK-T-2108, Type I, Class A. In the event of a conflict between the text of this specification and KKK-T-2107 or KKK-T-2108, this specification shall take precedence. The GVW ratings specified for a KKK-T-2107 Class C and KKK-T-2108 Class A truck chassis (19,000 and 24,000 pounds respectively) shall be the minimum acceptable. This GVW rating may be increased if required by the manufacturer's standard commercial (see 6.6) design. The following requirements shall be met in lieu of or in addition to the requirements of KKK-T-2107 and KKK-T-2108:

a. The vehicle engine shall operate on JP-8 fuel conforming to MIL-T-83133 without detrimental effect to its components. The engine shall be certified as having successfully completed NATO AEP-5 test procedures (400 hour dynamometer test) to assure JP-8 compatibility.

b. Dual controls shall be furnished to allow the operation of the vehicle from either side of the cab. All operating controls for chassis and sweeping unit, with the exception of the hopper dump gate control, shall be mounted inside the cab readily available to the operator in either right or left driver position, and shall include but not be limited to:

(1) Sweeper (auxiliary) engine ignition switch, throttle, tachometer, oil pressure gauge, water temperature gauge and hour meter.

(2) Sweeping/pickup unit controls.

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(3) Gutter broom controls and gutter broom light control.

(4) Controls for the dust suppression water system and low water warning device.

(5) All lighting controls; to include the rotating beacon or strobe light control.

c. Tire size shall be not less than 9.00X20-12 ply rating. When specified (see 6.2), the vehicle tires shall be inherently designed to provide resistance to ozone degradation with superior performance over paragraph 3.5.6 of MIL-T-12459. The tires shall be capable of meeting storage requirements IAW paragraph 3.5.5 of MIL-T-12459. When specified (see 6.2), the vehicle tires shall be coated with a rubber penetrating, preservative coating to provide additional protection against ozone degradation. Permissible preservative tire coatings shall penetrate into the outer layer of the rubber. Products which can be wiped or pulled off shall not be used. The following products, or equivalent, shall be used;

FSCM	Product Manufacturer	Product P/N
70228 -	Akron Paint & Varnish, Inc. P.O. Box 2765 Akron, OH 44301	LR-1588 (Grade A-1182)
62692 -	Chem-Pro Manufacturing Co, Inc. P.O. Box 213 Buffalo, NY 14221	ZR-2250
05568 -	UOP, Inc. (Universal Oil Products) P.O. Box 5017 Des Plaines, IL 60017-5017	UOP-288 or UOP-688

d. The rearview mirrors furnished shall also allow observation of either gutter broom and either end of the pickup head (or outboard ends of pickup heads on both the right and left-hand sides if more, than one pickup head is used) by the operator while seated in either operating position and facing forward.

e. An adjustable floodlight for each gutter broom shall be furnished. Also a single floodlight at rear of hopper, and a 2-lens 360° rotational amber beacon light or strobe light mounted atop the sweeper.

f. A slow moving vehicle sign or decal shall be furnished and mounted in a readily visible location on the back of the sweeper.

g. The sweeper shall be furnished with all special tools, if any, that are required for its operation together with the manufacturer's regular tool storage space or box.

3.3.2 Sweeping unit. The sweeping unit shall consist of at least the following components:

a. Sweeper power unit.

b. Fan.

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c. Sweeping/pickup unit.

d. Refuse separator.

e. Hopper.

f. Hydraulic system.

g. Gutter brooms.

h. Dust suppression system.

i. Hand suction hose.

3.3.2.1 Sweeper power unit. The sweeping unit shall be powered by a commercial (see 6.6), diesel, water cooled, industrial engine having sufficient rated power to meet the requirements specified herein. Engine shall be equipped with automatic shutdown devices against low oil pressure or high coolant temperatures. The engine shall have a replaceable type oil filter, electric starter, alternator with charge indicator light or gauge and a variable speed governor. The engine shall be furnished with oil pressure gauge, temperature gauge and hourmeter. The design of the engine air intake shall prohibit water, dust, etc. from entering the engine. The engine shall share the chassis engine fuel tank. The requirements of paragraph 3.3.1 subparagraph "a" shall apply to this engine also.

3.3.2.2 Fan. The fan shall be mounted on antifriction bearings. Design shall be conducive to long service and maintenance free life. The fan or blower shall create air pressure and suction or vacuum for the pickup head and hand suction hose. The fan or blower and its peripheral scroll shall be lined with replaceable abrasion resistant polymer material. Necessary valving and controls shall be included to permit use of either the pickup head or the hand suction hose.

3.3.2.2.1 Air volume. While operating at a rotational speed of 2,500* revolutions per minute (rpm) or within the fan manufacturer's recommended normal operating range, the fan shall generate not less than 7,000* standard cubic feet per minute (scfm)** of air, with -11.75 inches or greater water column suction static pressure. This pressure shall be generated at or as near as possible to the point at which the suction duct expands into the debris hopper and shall be measured with the sweeper in the operational mode.

*Acceptable alternate figures are 2,800 rpm and 7,500 scfm respectively.
**scfm is cfm converted to 72°F and 50 percent relative humidity.

3.3.2.3 Sweeping/pickup unit. The sweeping/pickup unit shall consist of either a pickup head or a combination of pickup heads and windrow broom. If used, the broom shall be hydrostatically driven and capable of pivoting for right or left hand sweeping. Broom fillers shall be polypropylene. The size I sweeper, excluding gutter brooms, shall provide a sweeping width of 78 inches minimum, perpendicular to the direction of travel, in either the right or left hand sweeping mode. The size II sweeper, excluding gutter brooms, shall provide a sweeping width of 87 inches minimum, perpendicular to the direction of travel, in either the right or left hand sweeping mode. Pickup head(s) shall include the necessary piping, hosing and fittings for connecting to the fan and hopper

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as necessary. The sweeping components shall be raised to the travel position and lowered to the work position hydraulically or pneumatically by controls in the truck cab. Ground clearance when in travel position shall be not less than 5 inches. If components are raised pneumatically, the system shall be completely independent of the chassis air system (i.e. separate compressor, air cylinders, air lines and reservoir tanks).

3.3.2.4 Refuse separator. A separator shall be installed in the system to remove the dirt and refuse from the air stream and allow same to be deposited in the hopper. Separator shall be designed such that it will not clog or plug with dirt and trash normally encountered, and shall be provided with a clean-out door(s). Hopper inspection door may be substituted as clean-out door if full-width inspection of separator is attained through the hopper door. Centrifugal separators, when included in the design, shall be equipped with liners and shall permit full-width inspection. Metal filter screens, when used, shall be 18-gauge minimum thickness.

3.3.2.5 Hopper. The hopper, which serves as a collection point for dust and refuse, shall be all steel welded construction with a minimum usable capacity of 6 cubic yards. The hopper shall be hydraulically dumped, through a hydraulic gate located at its rear, such that sweepings will be discharged behind the rear wheels. Design shall ensure that the body empties completely when in the open or raised position. One or more doors shall permit cleanout or inspection of refuse level in the hopper. Hopper shall be raised independently without the use of the auxiliary engine when necessary to allow access to the auxiliary engine for daily service and maintenance or for replacement.

3.3.2.6 Hydraulic system. The hydraulic system shall be compatible with the design requirements of the sweeper. The system shall consist of a pump, a reservoir, suction or pressure line strainer, replaceable type full-flow oil filter, necessary hydraulic cylinders, drive motors, control valves, relief valve, hoses, piping, standard fittings and an easily accessible hydraulic oil dipstick or easily visible sight gauge.

3.3.2.7 Gutter brooms. Hydrostatically driven gutter brooms, with steel or polypropelene fillers, shall be furnished and mounted on the right and left side of the sweeper. The brooms shall be capable of both lateral and vertical operation. The brooms shall be capable of extending a minimum of 18 inches beyond the sweeper's widest point. Brooms shall have angular adjustment and shall be capable of maintaining constant down pressures to obtain optimum sweeping performance on all surface contours. Brooms shall be designed with spring tension and relief valves to permit yielding from high force or impact, horizontally or vertically, and returning to their original position. Each broom operation shall be independently controlled by the driver in the cab.

3.3.2.8 Dust suppression system. The sweeper shall have a pressurized water system to suppress dust around the gutter brooms. The water system shall suppress dust around the pickup heads and inside the hopper, if needed, to control visible dust under normal sweeping operations. The system shall consist of a self-priming pump, spray nozzles for fogging action and sufficient water capacity for a minimum of 2 hours continuous operation with all nozzles spraying maximum amount. The system shall be provided with a low water warning device and an on/off valve between the water tank and outlets, controlled from the cab. A water hose, with storage rack, shall be provided for filling the water reservior. Controls for water valves shall be located in the operator's cab. The water pump shall be driven by the auxiliary engine. The dust suppression system shall satisfy all requirements of paragraph 4.6.4.7.

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3.3.2.9 Hand suction hose. A hand hose, at least 6 inches in diameter and long enough to pick up objects at an 8 foot distance from both sides of the vehicle, with nozzle, shall be furnished for cleaning areas inaccessible to normal operation. Design of the sweeper shall provide stowage space for this hand suction hose and nozzle when not in use.

3.4 Finish and protective coatings.

3.4.1 Painting. The entire exterior surface of the vehicle shall be painted with the manufacturer's standard commercial (see 6.6) polyurethane or acrylic urethane paint. The color of the finish coat shall be Forest Green, Color Number 24052 of FED-STD-595. These requirements shall supersede the painting requirements of KKK-T-2107 or KKK-T-2108.

3.4.1.2 Rustproofing. The level of rustproofing specified for the truck chassis in the KKK-T-2107 or KKK-T-2108 ordering data shall apply to the entire vehicle (i.e. to the truck chassis and the sweeping unit). The first article units shall not be rustproofed until after approval of the first article test report.

3.5 Markings. Markings shall be applied with black paint, Color Number 37038 of FED-STD-595.

3.5.1 Identification markings. The vehicle shall be marked as specified in KKK-T-2107 or KKK-T-2108. In addition, a transportation data plate shall be furnished as specified herein.

3.6 Identification plates and operating instructions.

3.6.1 Transportation data plate. A transportation data plate shall be provided. The plate shall contain at least the following information:

a. Side and rear silhouette views of the vehicle.

b. Horizontal and vertical location of the center of gravity of the vehicle in transport condition.

c. Shipping weight.

d. Loading cubage.

e. Overall length, width, and height.

f. Tiedown instructions for air transport in Types C-130, C-141, C-17 and C-5 Aircraft.

3.7 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.8 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.9 Fabrication.

3.9.1 Welders and welding.

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3.9.1.1 Welders. All welders employed in the fabrication of the vehicle, except for commercial (see 6.6) components, shall be certified in accordance with the applicable requirements of B2.1-84 Welding Qualifications of the American Welding Society .

3.9.1.2 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof and service loadings.

3.9.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Unless torque is specified, threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.9.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of an approved shape. Rivet heads shall be full, neatly made, concentric with the rivet holes and in full contact with the surface of the component.

3.9.4 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of backlash.

3.9.5 Cleaning. The vehicle shall be cleaned. Loose, spattered, or excess solder, welding slag, stray bolts, nuts and washers, rust, metal particles, pipe compound and other foreign matter shall be removed during and after final assembly.

3.10 Workmanship. The vehicle, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish and radii of fillets; thoroughness of soldering, welding, brazing, painting, wiring and riveting; marking of parts and assemblies; alignment of parts and tightness of assembly fasteners; etc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may use any inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that the vehicle conforms to the prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply

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with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the vehicle shall be classified as follows:

a. First article testing (see 4.4 and 6.4).

b. Acceptance testing (see 4.5).

4.3 Test conditions.

4.3.1 Apparatus. Apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at proper intervals to ensure laboratory accuracy.

4.3.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data and/or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Engine speed.

f. Engine oil pressure.

g. Engine coolant temperature.

h. Vehicle odometer reading at the start and finish of each test.

i. Engine hourmeter reading at the start and finish of each test.

j. Hydraulic fluid pressure at the pump outlet.

k. Hydraulic fluid temperature in the reservoir.

4.3.3 Preparation for tests. The vehicle, equipped with all specified equipment, shall be fully serviced and all equipment shall be prepared for immediate operation.

4.3.4 Test rejection criteria. Throughout all tests specified herein, the vehicle shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein.

b. Spillage or leakage of any liquid, including fuel, coolant, lubricant, or hydraulic fluid, under any operating condition, except as allowed herein.

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c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear.

e. Interference between vehicle components or between the vehicle, the ground, and all required obstacles, with the exception of normal contact by the tires.

f. Misalignment of components.

g. Evidence of undesirable roadability characteristics, including instability in handling during cornering and braking.

h. Conditions which present a safety hazard to personnel during operation, service, or maintenance.

i. Overheating of the engine, transmission, or any other vehicle component.

4.4 First article testing.

4.4.1 First article test sample. The first article test sample shall be identical to the items procured under the terms of the contract. Approval of the first article test report shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.4.2 First article tests. The first article tests shall consist of all tests listed under 4.6.

4.4.3 Test report. Upon completion of the first article tests, a first article test report shall be prepared in accordance with MIL-STD-831. An authorized Government representative shall countersign all data sheets and the report. The following shall be incorporated as a part of the test report:

a. All data collected, failures, maintenance performed, and other events recorded shall be identified by accumulated operation time, miles, cycles or position in the test procedure, as appropriate. Test conditions at the time of failures or irregular operations identified shall be recorded.

b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Summary of the engineering reasoning behind any corrections made, to be made or proposed to be made and the predicted effectiveness of those corrections.

d. List of special tools and diagnostic equipment necessary to perform field level maintenance.

4.5 Acceptance testing. Acceptance testing shall be classified as follows:

a. Individual testing (see 4.5.1).

b. Random sample testing (see 4.5.2).

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All defects identified shall be corrected prior to acceptance.

4.5.1 Individual testing. Each production vehicle shall be subjected to the tests specified in 4.6.1 and 4.6.2.

4.5.2 Random sample testing. Any production vehicle may be selected by the procuring activity at any time to be subjected to the tests described in 4.6.4. Upon completion of the random sample testing, the contractor shall prepare a test report in accordance with MIL-STD-831. An authorized Government representative shall countersign all data sheets of the report.

4.5.2.1 Random sample test failure. When a vehicle fails to successfully complete any random sample test, the cause of the failure shall be determined, corrective action shall be taken, and the test shall be repeated to verify adequacy of the corrective action. Tests not involving the defective component(s) may be continued pending the investigation of the failure. Vehicles to be delivered pursuant to the contract shall be inspected for all defects identified during all random sample tests and may not be accepted prior to verification of adequacy for the corrective action for a random sample test failure.

4.6 Test methods.

4.6.1 Examination of product. The vehicle shall be inspected to determine compliance with the requirements specified herein. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and their application, welding and riveting and markings. For the first article and random sample vehicles, this examination shall be accomplished by use of a checklist which lists each requirement not validated by a test and the results of the examination.

4.6.2 Preliminary operation. Sweepers produced in fulfillment of the contract or order shall be completely assembled, adjusted, lubricated and otherwise serviced for operation. A run-in test shall be conducted for each machine, before delivery, in accordance with the manufacturer's recommendations. During this operational test, all controls, adjusting mechanisms, brakes, hydraulic systems and other accessories shall be operated sufficiently to ensure against restrictions or malfunctions. Failure to pass any phase of this test shall be cause for rejection.

4.6.3 Mobility tests. The fully loader sweeper shall be driven on a paved road to determine compliance to 3.2.1.a. Additionally, the minimum right and left turning radii and cornering radii as well as the ground clearance shall be measured. Failure to meet the requirements of 3.2.1 subparagraphs b, c and d shall be cause for rejection.

4.6.4 Capacity testing.

4.6.4.1 Cleaning tests. Cleaning tests shall be conducted within an ambient temperature range of +32°F to +125°F. Test areas shall be thoroughly cleaned through the use of hand brooms and hand vacuum cleaners prior to placement of a weighed amount of materials for pickup tests. The requirements of each individual cleaning test specified herein shall be met by one pass of the sweeper over the test area at the stipulated speed. If the sweeper wets the test area, through the use of water during a cleaning test run, then the test area shall be allowed to dry after the sweeping pass. After the area has dried, the material remaining thereon shall be picked up through the use of hand brooms

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and hand vacuum cleaners. The picked up material shall then be weighed so that the percentage of cleaned matter retained by the vehicle can be determined. All tests shall be conducted without the use of gutter brooms, except the street pickup test.

4.6.4.2 Sand pickup test. Commercially procured dry sand shall be spread uniformly over paved, dry, clean surfaces, 5 feet wide and 60 feet long. Using the procedures specified above, the areas shall be cleaned and the percentage of pickup shall be determined. Failure of the sweeper to meet the provisions of 3.2.2.1 shall be cause for rejection.

4.6.4.3 Gravel pickup test. Pea gravel shall be uniformly spread over two cleaned, paved areas each measuring 5 feet wide and 60 feet long. Using the procedures described above, the areas shall be cleaned and the percentage of pickup determined. Failure to meet the provisions of 3.2.2.2 shall be cause for rejection.

4.6.4.4 Stone pickup test. Twenty stones shall be placed on a cleaned, paved surface, in a line approximately 36 inches apart. The sweeper shall make a 3 mph (minimum) pass over the stones. Failure to meet the criteria specified in 3.2.2.3 shall be cause for rejection.

4.6.4.5 Street pickup test. Thirty each of the items listed in 3.2.2.4 shall be alternately dispersed in the street gutter line, including the 15 foot radius curve, approximately 1-1/2 feet apart. The sweeper shall, on a single pass, at 3 mph (minimum), pick up and retain 100 percent of this general trash.

4.6.4.6 Hand hose test. The ability of the hand hose to pick up and retain at least one of each of the items listed in 3.2.2.5, randomly placed at distances up to 8 feet from either side of the cleaner, shall be demonstrated. Failure to pick up and retain 100 percent of these items shall be cause for rejection.

4.6.4.7 Dust emission test. Those sweepers designed to exhaust to the surrounding air shall be observed from a distance of 50 feet during tests 4.6.4.2 through 4.6.4.6. Visible emissions from the sweepers shall be cause for rejection.

4.6.4.8 Wet cleaning test. An area of pavement measuring 10 feet wide and 100 feet long shall be hosed until water is standing on the entire area. The sweeper shall then clean the wet area at 3 mph minimum. Any harmful or adverse effects upon the filter system or failure to pick up the standing water shall be cause for rejection.

4.6.4.9 Air volume test. The air volume shall be measured at or as near as possible the point at which the suction duct expands into the debris hopper, while the sweeper is operating under the conditions specified in 3.3.2.2.1. This measurement shall be taken with a commercial flow cross metering system that has been calibrated against an Air Movement and Conditioning Association (AMCA)* flow standard. Air volume shall then be obtained through use of calibration charts showing the calibration between the in-line flow station and the AMCA standard. Failure to meet the scfm quantity specified in 3.3.2.2.1 shall be cause for rejection. In lieu of the foregoing, the manufacturer, at his option, may furnish a notarized certification supported by engineering data, (that is certified by a registered professional engineer) that the scfm of the sweeper being furnished either meets or exceeds the scfm specified in 3.3.2.2.1 under the conditions stipulated therein.

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*Address: Air Moving and Conditioning Association (AMCA)
30 West University Drive
Arlington Heights IL 60004

4.6.5 Design verification.

4.6.5.1 Environmental test. The supplier shall furnish certification that the sweeper will either meet or exceed the requirements of 3.1.4. Failure to do so shall be cause for rejection.

4.6.5.2 Weight and weight distribution. The weight on each corner of the vehicle shall be measured with the unit standing on a flat level surface, both in the empty and fully loaded conditions, to demonstrate compliance with 3.1.4. The fully loaded vehicle shall include ballast to simulate a 250 pound crew member in each seating position.

4.6.5.3 Electromagnetic interference measurement. The manufacturer shall submit notarized certification that the vehicle complies with the requirements of 3.1.5.

4.6.5.4 Sound level testing.

4.6.5.4.1 Cab interior sound level test. The vehicle shall be tested to determine conformance to the cab interior sound level requirement of 3.1.6.1. The test criteria and parameters shall be as specified in paragraphs 5.1.2 through 5.1.5 of MIL-STD-1474, except that the vehicle shall be in a sweeping mode and operated at a forward speed of not less than 6 mph in lieu of the requirements of paragraph 5.1.2.3.1.1 of MIL-STD-1474.

4.6.5.4.2 Exterior sound level test. The vehicle shall be tested to determine conformance to the exterior sound level requirement of 3.1.6.2. The sound level shall be measured in accordance with the criteria and parameters of paragraphs 5.3.1 through 5.3.2 of MIL-STD-1474, except that the vehicle shall be in a sweeping mode and operated at a forward speed of not less than 6 mph throughout the measurement zone in lieu of the test procedure described in SAE J366.

4.6.5.5 Air transportability analysis. A detailed engineering analysis shall be conducted to demonstrate compliance with 3.1.8. The analysis shall include the tiedowns and all major components and their ability to withstand the accelerations specified in MIL-STD-1791. The evaluation shall also include a dimensional analysis for the vehicle while loaded aboard Types C-130, C-141, C-17 and C-5 Aircraft.

4.6.5.6 Tire ozone resistance test. When ozone resistant tires are required (see 3.3.1, subparagraph c), the components and materials used in the construction of the tires shall be subjected to, or certified as having passed, or being capable of passing the tests in paragraph 4.7.9 of MIL-T-12459 to demonstrate compliance with paragraph 3.3.1, subparagraph c of this purchase description. If the tires cannot be certified as capable of passing the required tests, the tire material shall be tested in accordance with paragraphs 4.7.9 through 4.7.9.9 of MIL-T-12459 to demonstrate compliance with paragraph 3.3.1, subparagraph c of this purchase description. In addition, identically made stored tires shall be examined in accordance with paragraph 4.7.8 of MIL-T-12459, except the term "extended period" shall refer to a minimum of three years.

5. PREPARATION FOR DELIVERY

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5.1 Preparation for delivery shall be in accordance with the terms of the contract.

6. NOTES

6.1 Intended use. The sweeper covered by this specification is intended for the dual use of cleaning streets and airfield areas such as runways, taxiways, parking ramps and other airfield paved areas required for aircraft operations.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type sweeper required (see 1.2).

c. Any ordering data as required by KKK-T-2107 or KKK-T-2108 (see 3.3.1).

d. If ozone resistant tires are required (for SWA vehicles only, see 3.3.1, subparagraph c).

e. If ozone resistant tire coating is required (for WRM vehicles, see 3.3.1, subparagraph c).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Subject term (key word) listing.

Sweeper
Vacuum
Blower
Chassis
Engine
Hydraulic Systems
Performance

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6.6 Definitions.

6.6.1 Commercial. A manufacturer's standard (product) having been marketed, in production and sold in substantial quantities to the general public for a minimum of one year prior to the date of invitation for bid or request for proposal. An item is "sold in substantial quantities" only when the quantities regularly sold are sufficient to constitute a real commercial market. Nominal quantities such as models, samples, prototypes or experimental units do not meet this requirement. The "general public" is a significant number of buyers other than the Government or affiliates of the offeror; the item involved must not be for Government end use. For the purpose of this specification items acquired for "Government end use" include items acquired for foreign military sales.

MILITARY INTERESTS:

Custodians

Air Force - 99

Review Activities

Navy - YD

PREPARING ACTIVITY:

Air Force - 84

Project 3825-0214

FSC 3825

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